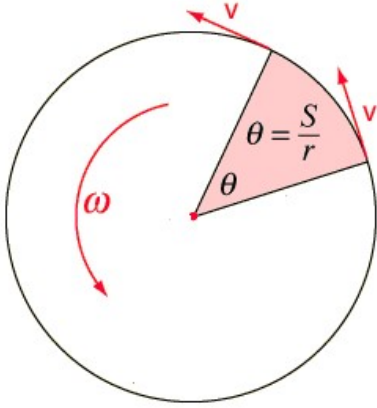




ধরি, T সময়ে অতিক্রান্ত দূরত্বের পরিমাণ = $2\pi r$

$$\therefore \text{একক ,, ,, ,, ,,} = \frac{2\pi r}{T}$$

$$\therefore \text{Linear Velocity } v = \frac{2\pi r}{T} \text{-----(1)}$$

আবার যদি Angular velocity = ω হয়, তাহলে

T সময়ে অতিক্রান্ত কোণের পরিমাণ = 2π

$$\therefore \text{একক ,, ,, ,, ,,} = \frac{2\pi}{T}$$

$$\therefore \text{Angular Velocity } \omega = \frac{2\pi}{T} \text{-----(2)}$$

From (1),

$$v = \frac{2\pi r}{T} = r \frac{2\pi}{T} = r\omega \text{ [from (2)]-----(3)}$$

Frequency: T সময়ে পূর্ণ দোলন সম্পন্ন হয় ১ টি

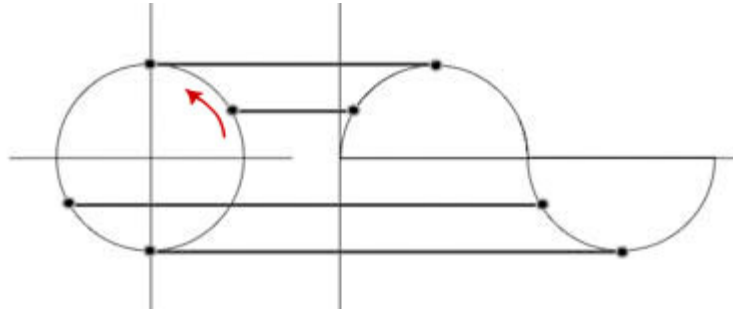
$$\therefore \text{একক সময়ে -----} = \frac{1}{T}$$

$$\therefore \text{Frequency (f) = } \frac{1}{T} \text{ Cycle/ sec [Here T is period]}$$

$$\therefore T = \frac{1}{f}$$

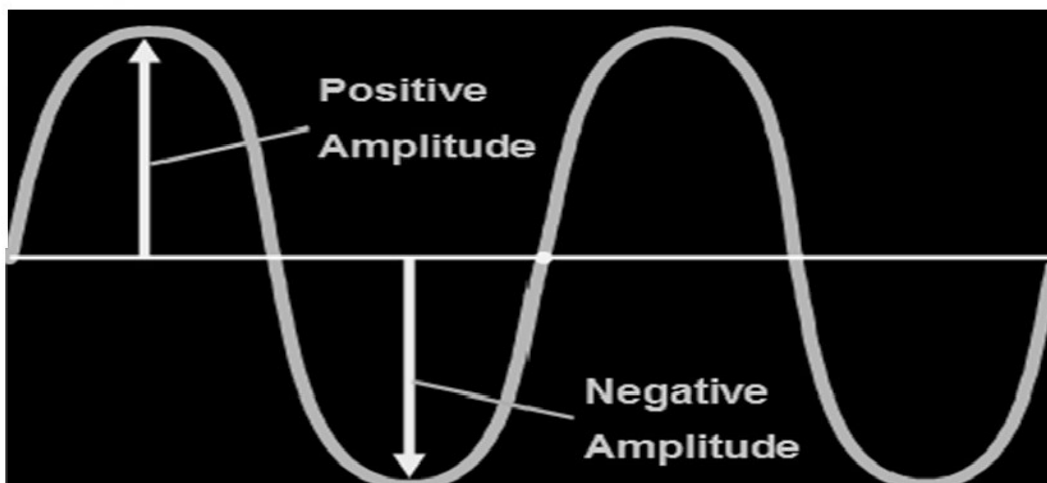
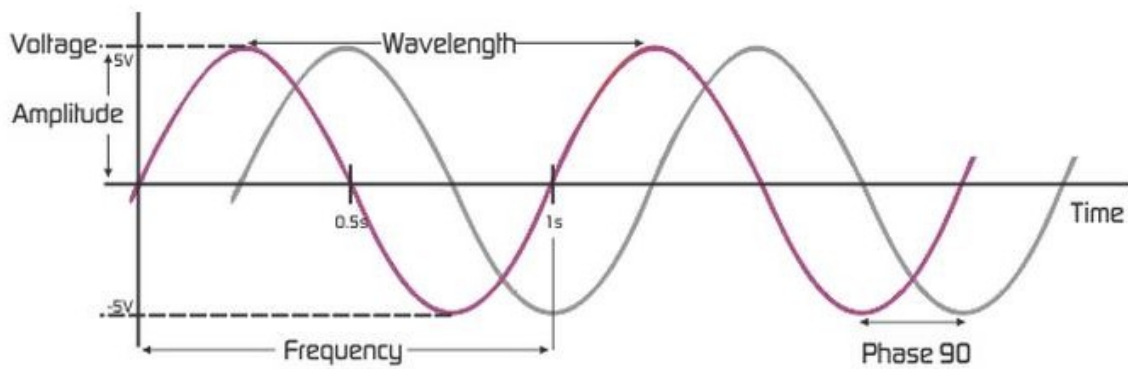
$$\therefore \text{Angular Velocity } \omega = \frac{2\pi}{T} = \frac{2\pi}{\frac{1}{f}} = 2\pi f$$

One rotation around the circle completes one cycle of rising and falling in the wave, as seen in the picture below.



Waves are familiar to us from the ocean, the study of sound, [earthquakes](#), and other natural phenomenon. But as any surfer can tell you, ocean waves come in very different sizes, as can all waves. To fully understand waves, we need to understand measurements associated with these waves, such as how often they repeat (their [frequency](#)), and how long they are (their [wavelength](#)), and their vertical size (amplitude).

While these measurements help describe waves, they do not help us make predictions about wave behavior. In order to do that, we need to look at waves more abstractly, which we can do using a mathematical formula. It is possible to look at waves mathematically because a wave's shape repeats itself over a consistent interval of time and distance. This behavior mirrors the repetition of the circle. Imagine drawing a circle on a piece of paper. Now imagine drawing that same shape while your friend slowly pulled the piece of paper out from under your pencil – the line you would have drawn traces out the shape of a wave. To better appreciate this idea, take a look at the "Circle to a Wave" link in the *Experiment!* section on the right menu. One rotation around the circle completes one cycle of rising and falling in the wave, as seen in the picture below



Q-1: Draw the functions

$$y = f(x) = x; \quad -2 \leq x \leq 2;$$

$$= 4 - x; \quad 2 \leq x \leq 4;$$

```
x1=-2:2;
y1=x1
x2=2:4;
y2=4-x2;
x=[x1 x2];
y=[y1 y2]
plot(x,y);
```

Q-2: Draw the functions $y = A \sin \omega t$ with frequency one, Amplitude one

```
A=1; % A means amplitude
t=0:.001:1; % t = 0 to 1 means time in one second
f=1; % frequency = 1
w=2*pi*f;
y=A*sin(w*t);
plot (t,y)
```

Q-3: Draw the functions $y = A \sin \omega t$ with frequency two, Amplitude one and number of frequency (cycle) after 1 second

```
A=1; % A means amplitude
t=0:.001:1;
f=2; % frequency = 2
w=2*pi*f;
y=A*sin(w*t);
plot(t,y)
```

Q-4: Draw the functions $y = A \sin \omega t$ with frequency one, Amplitude four and number of frequency (cycle) after 1 second

```
A=4; % A means amplitude
t=0:.001:1;
f=1; % frequency = 1
w=2*pi*f;
y=A*sin(w*t);
plot(t,y)
```

Q-5: Draw the functions $y = A \sin \omega t$ with frequency one, Amplitude four and number of frequency after 5 seconds

```
A=4; % A means amplitude
t=0:.001:5;
f=1; % frequency = 1
w=2*pi*f;
y=A*sin(w*t);
plot(t,y)
```

Q-6: Draw the functions $y = A \sin \omega t$ with frequency two, Amplitude two and number of frequency after 5 seconds

```
A=2; % A means amplitude
t=0:.001:5;
f=2; % frequency = 2
w=2*pi*f;
y=A*sin(w*t);
plot(t,y)
```

Q-7: Draw the functions $y = A \cos \omega t$ with frequency one, Amplitude one and number of frequency after 1 seconds

```
A=1; % A means amplitude
t=0:.001:1;
f=1; % frequency = 1
w=2*pi*f;
y=A*cos(w*t);
plot(t,y)
```

Q-8: Draw the functions $y = A \cos \omega t$ with frequency three, Amplitude one and number of frequency after 2 seconds

```
A=1; % A means amplitude  
t=0:.001:2;  
f=3; % frequency = 3  
w=2*pi*f;  
y=A*cos(w*t);  
plot(t,y)
```

Q-9: Draw the functions $y = A \sin(\omega t + b)$ with frequency one, Amplitude one and number of frequency one after 1 seconds with 90° phase shift

```
A=1; % A means amplitude  
t=0:.001:1;  
f=1; % frequency = 1  
w=2*pi*f;  
b=0;  
y=A*sin(w*t+b);  
plot(t,y)
```

Q-10: Draw the functions $y = A \sin(\omega t + b)$ with frequency one, Amplitude one and number of frequency one after 1 seconds with 0° and 90° phase shift

```
%Let  
% y=A*sin(w*t+b);  
% z=A*sin(w*t+c);
```

$A=1$; *% A means amplitude*

$t=-0.5:0.001:1$;

$f=1$; *% frequency = 1*

$w=2 \times \pi \times f$;

$b=0$;

$c=\pi/2$;

$y=A \sin(w \times t + b)$;

$z=A \sin(w \times t + c)$;

$\text{plot}(t,y,t,z)$

Q-11: Draw the functions $y = A \sin(\omega t + b)$ with frequency one, Amplitude one and number of frequency one after 1 seconds with 0° and 90° phase shift

```
%Let  
% y=A*sin(w*t+b);  
% z=A*sin(w*t+c);
```

$A=1$; *% A means amplitude*

$t=-0.5:0.001:1$;

$f=1$; *% frequency = 1*

$w=2 \times \pi \times 1$; *% $\omega = 2\pi f$*

$b=0$;

$c=\pi/2$;

```
y=A*sin(w*t+b);
z=A*sin(w*t+c);
plot(t,y,t,z)
```

Q-12: Draw the functions $y = A \cos \omega t$ with frequency five, Amplitude one and number of frequency after 1 seconds with 45° phase shift

```
A=1; % A means amplitude
t=0:.001:1;
w=2*pi*5; %  $\omega = 2\pi f$ 
b=pi/4;
y=A*cos(w*t+b);
plot(t,y)
```

Q-13: Draw the functions $y = A \cos \omega t$ with frequency five, Amplitude one and number of frequency after 1 seconds with 60° phase shift.

[If the phase shift is positive, the horizontal shift is to the right. If the phase shift is negative, the horizontal shift is to the left]

```
A=1; % A means amplitude
t=0:.001:1;
w=2*pi*5; %  $\omega = 2\pi f$ 
b=pi/3;
y=A*cos(w*t+b);
plot(t,y)
```

Q-14: Draw the functions $y = A \cos \omega t$ with frequency five, Amplitude one and number of frequency after 1 seconds with 60° phase shift.

```
A=1; % A means amplitude
t=0:.001:1;
w=2*pi*5; %  $\omega = 2\pi f$ 
b=pi/3;
y=A*cos(w*t - b);
plot(t,y)
```

Q-15: Draw the functions $y = A \cos \omega t$ with frequency three, Amplitude one and number of frequency after 1 second with asterisk (*)

```
A=1; % A means amplitude
t=0:0.01:1;
w=2*pi*3; %  $\omega = 2\pi f$ 
cosine=A*cos(w*t);
plot(t,cosine,'*-')
```

Q-16: Draw the functions $y = A \cos \omega t$ with frequency three, Amplitude one and number of frequency after 1 second with asterisk (+)

```

A=1; % A means amplitude
t=0:0.01:1;
w=2*pi*3; %  $\omega = 2\pi f$ 
cosine=A*cos(w*t);
plot(t,cosine,'+')

```

Q-17: Draw the functions $y = A \cos \omega t$ with frequency three, Amplitude one and number of frequency after 1 second with asterisk (*) and red color

```

A=1; % A means amplitude
t=0:0.01:1;
w=2*pi*3; %  $\omega = 2\pi f$ 
cosine=A*cos(w*t);
plot(t,cosine,'r+')

```

Question: A spring for which $k = 700 \text{ Nm}^{-1}$ hangs in a vertical position with its upper end fixed. A mass of 7 kg is attached to the lower end. After coming to rest, the mass is pulled down 0.05 m and released. Discuss the resulting motion of the mass, neglecting air resistance.

[একটি 7kg বস্তুকে স্প্রিং এর শেষ প্রান্তে ঝুলিয়ে দেওয়া হল ফলে স্প্রিং টি কিছুক্ষণ উপরে নীচে দুলতে থাকবে, এক সময় স্থির হয়ে যাবে। আমি যদি ০.০৫ মিটার নিচের দিকে টেনে আবার ছেড়ে দেই তাহলে উপরে নীচে দুলতে থাকবে, বিভিন্ন সময়ে এই উঠানামর পরিমাণ কত?]

Answer: Let x , positive when measured downward be the change in position of the mass at time t . When the mass is at rest the spring force is equal to but opposite in direction to the gravitational force. The net force at time t is the spring force $-kx$ corresponding to the change x in the position of the mass. Then

We have,

$$\frac{d^2 x}{dt^2} + \frac{k}{m}x = 0$$

$$\frac{d^2 x}{dt^2} + \frac{700}{7}x = 0$$

$$\frac{d^2 x}{dt^2} + 100x = 0 \text{-----(1)}$$

Auxiliary Equation is:

$$m^2 + 100 = 0$$

$$m^2 = -100$$

$$m = \pm \sqrt{-100}$$

$$m = \pm i\sqrt{100}$$

$$m = \pm i\sqrt{100}$$

$m = \pm i10$ [As like as $\alpha \pm i\beta$]

So here $\alpha = 0$ and $\beta = 10$

Hence the general solution is: $x = e^{\alpha t} (A \cos \beta t + B \sin \beta t)$

$$x = e^{0 \cdot t} (A \cos 10t + B \sin 10t)$$

$$x = (A \cos 10t + B \sin 10t) \text{-----}(2)$$

Differentiating (2) with respect to t , we get,

$$\text{Velocity, } v = \frac{dx}{dt} = -A \sin 10t \times (10) + B \cos 10t \times (10)$$

$$v = \frac{dx}{dt} = -10A \sin 10t + 10B \cos 10t \text{-----}(3)$$

When $t = 0$, $x = 0.05$ and $v = 0$,

Then from (3), we get,

$$0 = -10A \sin 10 \times 0 + 10B \cos 10 \times 0$$

$$0 = 10B \cos 10 \times 0$$

$$0 = 10B \cos 0$$

$$0 = 10B \Rightarrow B = 0$$

Again from (2), we get,

$$0.05 = A \cos 10 \times 0 + B \sin 10 \times 0$$

$$0.05 = A \cos 0 + B \sin 0$$

$$0.05 = A \cos 0 \Rightarrow A = 0.05$$

Hence the exact solution is: $x = 0.05 \cos 10t + 0 \times \sin 10t$

$$x = 0.05 \cos 10t \text{ Answer.-----}(4)$$

যদি t সময়ে কোনের displacement এর পরিমাণ হয় $=\theta$

$$\text{একক} \quad \text{,,} \quad \text{,,} \quad \text{,,} \quad \frac{\theta}{t}$$

$$\text{Angular velocity } \omega = \frac{\theta}{t}$$

$$\Rightarrow \theta = \omega t \text{-----}(\text{৫})$$

$$\text{Again we have, } \omega = \frac{2\pi}{T}$$

$$T = \frac{2\pi}{\omega} \text{-----}(6)$$

So, from (4) and (5), we can say $\Rightarrow \theta = 10t$ [Here $\omega = 10$]

From (6), we can say $T = \frac{2\pi}{10}$

So the resulting motion of the mass is:

i) $x = 0.05 \cos 10t$

ii) The period is $T = \frac{2\pi}{10}$ sec

iii) The frequency is $\frac{1}{T} = \frac{1}{\frac{2\pi}{10}} = \frac{10}{2\pi}$ cycles/sec

iv) The amplitude is 0.05 m

Q-18: Now Draw the Function Using MATLAB:

$$x = 0.05 \cos 10t$$

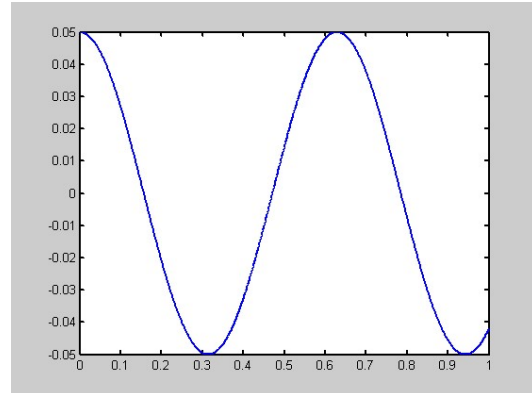
Here

$$A = 0.05$$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{\frac{1}{f}} = 2\pi f = 2\pi \times \frac{10}{2\pi}$$

Code:

```
A=0.05;  
t=0:.001:1;  
w=2*pi*10/(2*pi);  
y=A*cos(w*t);  
plot(t,y)
```



Q: An 8-lb weight is placed upon the lower end of a coil spring suspended from the ceiling. The weight comes to rest in its equilibrium position, thereby stretching the spring 6 inch. The weight is then pulled down 3 inch below its equilibrium position and released at $t = 0$ with an initial velocity of 1 ft/sec, directed downward. Neglecting the resistance of the medium and assuming that no external forces are present, determine the amplitude, period and frequency of the resulting motion. [By Hooke's Law $F = ks$]

Answer:

Here, $F = 8$ lb

$$m = \frac{W}{g} = \frac{8}{32} = \frac{1}{4}$$

Since the 8-lb weight stretches the spring 6 inch, i.e. $s = 6 \text{ inch} = \frac{6}{12} \text{ ft} = \frac{1}{2} \text{ ft}$

By Hook's Law,

$$F = ks$$

$$\therefore \text{String constant, } k = \frac{F}{s} = \frac{8}{\frac{1}{2}} = 16$$

We have,

$$m \frac{d^2 x}{dt^2} + kx = 0$$

$$\frac{1}{4} \frac{d^2 x}{dt^2} + 16x = 0$$

$$\frac{d^2 x}{dt^2} + 64x = 0 \text{ -----(1)}$$

Auxiliary Equation is: $m^2 + 64 = 0$

$$m = \pm \sqrt{-64}$$

$$m = \pm \sqrt{i^2 64}$$

$$m = \pm i \sqrt{64} [\alpha \pm i\beta]$$

$$m = \pm i8$$

Hence the general solution is: $x = e^{\alpha t} (A \cos \beta t + B \sin \beta t)$

$$x = e^{0 \times t} (A \cos 8t + B \sin 8t)$$

$$x = A \cos 8t + B \sin 8t \text{ -----(2)}$$

Since the weight is pulled down 3 inch = $\frac{3}{12} = \frac{1}{4} \text{ ft}$ below its equilibrium position and released at $t = 0$ with an initial velocity of 1 ft/sec, directed downward.

Hence according to question, the initial condition is:

$$x = \frac{1}{4} \text{ when } t = 0 \text{ i.e. } x(0) = \frac{1}{4} \text{ -----(3)}$$

and

$$v = \frac{dx}{dt} = x'(t) = 1 \text{ When } t = 0 \text{ i.e. } x'(0) = 1 \text{ -----(4)}$$

From (2),

$$x = A \cos 8t + B \sin 8t$$

$$\therefore \frac{dx}{dt} = -8A \sin 8t + 8B \cos 8t \text{ -----(5)}$$

$$\therefore 1 = -8A \sin 8 \times 0 + 8B \cos 8 \times 0 \text{ [From (4)]}$$

$$8B \cos 0 = 1 \therefore B = \frac{1}{8}$$

From (2),

$$x = A \cos 8t + B \sin 8t$$

$$\frac{1}{4} = A \cos 8 \times 0 + B \sin 8 \times 0 \text{ [From (3)]}$$

$$A = \frac{1}{4}$$

Hence the exact solution is:

$$x = \frac{1}{4} \cos 8t + \frac{1}{8} \sin 8t \text{ -----(6) Answer}$$

$$\text{Angular velocity } \omega = \frac{\theta}{t}$$

$$\Rightarrow \theta = \omega t \text{ -----(7)}$$

$$\text{Again we have, } \omega = \frac{2\pi}{T}$$

$$T = \frac{2\pi}{\omega} \text{ -----(8)}$$

So, from (7), we can say $\Rightarrow \theta = 8t$ [Here from (6) $\omega = 8$]

$$\text{From (8), we can say } T = \frac{2\pi}{8} = \frac{\pi}{4} \text{ sec}$$

$$\therefore \text{The period is } T = \frac{\pi}{4} \text{ sec and}$$

$$\text{The frequency is } \frac{1}{T} = \frac{1}{\frac{\pi}{4}} = \frac{4}{\pi} \text{ cycles/sec and}$$

$$\text{The amplitude is } \sqrt{\left(\frac{1}{4}\right)^2 + \left(\frac{1}{8}\right)^2} = \sqrt{\frac{1}{16} + \frac{1}{64}} = \sqrt{\frac{5}{64}} = \frac{\sqrt{5}}{8} \approx .28 \text{ ft [From (6)]}$$

Answer

So the resulting motion of the mass is:

v) $x = \frac{1}{4}\cos 8t + \frac{1}{8}\sin 8t$ Where 1st Amplitude is $\frac{1}{4}$ and the second amplitude is $\frac{1}{8}$

vi) The period is $T = \frac{\pi}{4}$ sec

vii) the frequency is $\frac{1}{T} = \frac{1}{\frac{\pi}{4}} = \frac{4}{\pi}$ cycles/sec and

viii) Final amplitude is 0.28 ft

RMS Amplitude

Note that unlike the sinusoid it makes little sense to talk about a peak amplitude when describing gaussian noise. This is because there are many peak amplitudes and any one is not representative of the amplitude of the noise overall. How should we measure the amplitude of a gaussian noise? We can't take the average of the instantaneous amplitudes since negative and positive values always cancel yielding an average value of zero. To prevent this we square each amplitude making all values positive. We then take the square root of the average of the squared values to get back to an amplitude measure. The resultant measure is referred to as the root-mean-square or RMS amplitude. The RMS amplitude is a more commonly accepted measure of amplitude because it applies to periodic sounds like sinusoids as well as to noise. The computational formula for RMS is

$$RMS = \sqrt{\frac{\sum p^2}{N}}$$

Where, p refers to an instantaneous amplitude or pressure value, and N refers to the number of these values in the time waveform.

Q-19: Now Draw the Function Using MATLAB:

$$x = \frac{1}{4}\cos 8t + \frac{1}{8}\sin 8t$$

Here

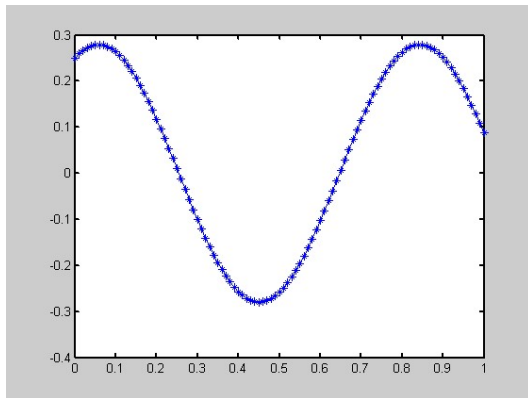
$$A = \sqrt{\left(\frac{1}{4}\right)^2 + \left(\frac{1}{8}\right)^2} = \sqrt{\frac{1}{16} + \frac{1}{64}} = \sqrt{\frac{5}{64}} = \frac{\sqrt{5}}{8} \approx .28 \text{ ft}$$

$$\omega = 8$$

Code:

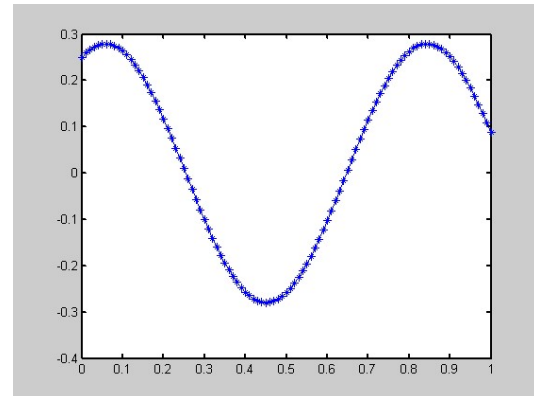
```
A1=1/4;B1=1/8;
t=0:0.01:1;
w=8;
y=A1*cos(w*t);
```

```
z=B1*sin(w*t);
x=y+z
plot(t,x)
```



OR

```
A1=1/4;B1=1/8;
t=0:0.01:1;
w=2*pi*4/pi;
y=A1*cos(2*pi*4/pi*t);
z=B1*sin(2*pi*4/pi*t);
x=y+z
plot(t,x)
```



Q-20: Now Draw the Function Using MATLAB within same axes:

i) $x = \frac{1}{4} \cos 8t$

ii) $x = \frac{1}{8} \sin 8t$

iii) $x = \frac{1}{4} \cos 8t + \frac{1}{8} \sin 8t$

Code:

```
A1=1/4; B1=1/8;
t=0:0.01:1;
w=8;
y=A1*cos(w*t);
z=B1*sin (w*t);
x=y+z
plot (t,y,t,z,t,x)
```

